

CIS 203: Capstone Project: Relational Database Design and Business Intelligence Dashboard

I. Project Objective

The primary objective of this capstone project is to provide students with hands-on experience in designing, implementing, and managing a relational database using MySQL, and subsequently leveraging that data for business intelligence through PowerBI. Students will choose a real-world application, demonstrating their ability to translate conceptual requirements into a robust database schema and derive meaningful insights through interactive dashboards.

Specific Learning Objectives:

- Database Design: Develop a strong understanding of relational database principles, including normalization, entity-relationship modeling, and data integrity.
- MySQL Proficiency: Gain practical experience in creating databases, tables, defining relationships, inserting data, and writing complex queries using MySQL.
- Data Modeling: Translate real-world scenarios into an effective ERD, identifying entities, attributes, and relationships.
- SQL Mastery: Execute various SQL commands, including DDL (CREATE, ALTER, DROP) and DML (INSERT, UPDATE, DELETE, SELECT), with emphasis on joins and subqueries.
- Business Intelligence (BI) Integration: Learn to connect PowerBI to a MySQL database and import data for analysis.
- Dashboard Development: Design and create interactive PowerBI dashboards that effectively visualize data, answer business questions, and provide actionable insights.
- Problem-Solving & Critical Thinking: Apply database and BI concepts to solve practical data management and analysis challenges.
- Project Management & Documentation: Plan, execute, and document a multi-faceted project, including database schema, query examples, and dashboard design.

II. Project Deliverables

Students will be required to submit the following:

1. Project Proposal: A brief document outlining their chosen theme, a high-level overview of the intended database, and potential insights they aim to extract.
2. A Project Summary Worksheet with screenshots of 3, 4, 5, 6 and 7.
3. ERD Diagram: A clear and well-structured Entity-Relationship Diagram for their chosen theme. Screenshots will be added to the Project Summary Worksheet.
4. MySQL Workbench Model: The .mwb file generated from MySQL Workbench, showcasing the complete database schema. A screenshot of the .mwb file will be added to the Project Summary Worksheet.

5. SQL Script: A single .sql file containing all DDL (CREATE TABLE, ALTER TABLE) and DML (INSERT) statements necessary to recreate their database and populate it with sample data. A screenshot of the .sql file will be added to the Project Summary Worksheet.
6. Query Documentation: A document detailing at least 10 meaningful queries, including examples of various join types (INNER, LEFT, RIGHT, FULL OUTER - emulated if not directly supported by MySQL), subqueries, and aggregate functions. Each query should have a clear explanation of its purpose. This will be added to the Project Summary Worksheet.
7. PowerBI Dashboard: A .pbix file containing a well-designed and interactive dashboard that effectively visualizes data from their MySQL database. This will be added to the Project Summary Worksheet.
8. Project Presentation: A comprehensive presentation summarizing their project, including:
 - Introduction to the chosen theme and its relevance.
 - Database design rationale.
 - Challenges faced and solutions implemented.
 - Key insights derived from the PowerBI dashboard.
 - Future enhancements.

III. Project Planning & Detailed Requirements

Phase 1: Theme Selection & Conceptual Design (Day 1)

- Theme Choice: Students will select a real-world theme of their choice. Examples include:
 - E-commerce Store Management
 - Library Management System
 - University Course Registration System
 - Hospital Patient Management
 - Inventory Management for a Small Business
 - Sports League Management
 - Movie Rental Database
 - Personal Finance Tracker
- Brainstorming Entities & Attributes: Identify the core entities within their chosen theme and their relevant attributes.
- Initial Relationship Identification: Begin to think about how these entities relate to each other.
- Project Proposal Submission.

Phase 2: ERD Diagram & Logical Design (Days 1)

- Develop the ERD Diagram:
 - Use a suitable ERD modeling tool (e.g., MySQL Workbench, draw.io, Lucidchart).
 - Clearly identify all entities.

- Define primary keys and foreign keys for each entity.
- Specify attributes for each entity, including data types and constraints (e.g., NOT NULL, UNIQUE).
- Illustrate relationships between entities, including cardinality (one-to-one, one-to-many, many-to-many) and optionality.
- Deliverable: ERD Diagram (image file or link).

Phase 3: MySQL Database Implementation (Day 2)

- MySQL Workbench Model Creation:
 - Translate the ERD into a physical database model within MySQL Workbench.
 - Create tables, define columns with appropriate data types, and set up primary and foreign key constraints.
 - Deliverable: .mwb file.
- SQL Script Generation:
 - Generate a SQL script from MySQL Workbench that creates the entire database schema (or write it manually based on the ERD).
 - Populate with Sample Data: Insert a meaningful amount of realistic sample data into all tables (at least 10-15 rows per table where applicable, more for larger tables). Ensure data integrity constraints are met.
 - Deliverable: SQL Script (.sql file).

Phase 4: Advanced SQL Queries (Day 2)

- Query Development: Write at least 10 complex and meaningful SQL queries that demonstrate a strong understanding of database querying. These should include:
 - SELECT statements with WHERE clauses.
 - Aggregate functions (COUNT, SUM, AVG, MIN, MAX) with GROUP BY and HAVING clauses.
 - Various Join Types:
 - INNER JOIN: At least two examples.
 - LEFT JOIN: At least two examples.
 - RIGHT JOIN: At least two examples.
 - FULL OUTER JOIN (emulated if necessary): Show how to combine LEFT and RIGHT joins with UNION to achieve a full outer join.
 - Subqueries: At least two examples (e.g., in WHERE, FROM, or SELECT clauses).
 - ORDER BY and LIMIT clauses.
 - Demonstrate understanding of data retrieval for specific business questions related to their chosen theme.
- Documentation: For each query, provide:

- The SQL code.
- A clear English description of what the query retrieves and why it's useful.
- Expected output (optional, but good practice).
- Deliverable: Query Documentation.

Phase 5: PowerBI Integration & Dashboard Development (Day 3)

- Connect PowerBI to MySQL:
 - Successfully establish a connection between PowerBI Desktop and their MySQL database.
 - Import relevant tables for analysis.
- Data Transformation (if necessary):
 - Use Power Query Editor to clean, transform, and shape data as needed for reporting (e.g., changing data types, creating calculated columns, merging queries).
- Dashboard Design:
 - Create a visually appealing and interactive PowerBI dashboard.
 - Include a variety of visualizations (e.g., bar charts, line charts, pie charts, tables, cards, maps - if applicable) that effectively tell a story about their data.
 - Key Performance Indicators (KPIs): Display relevant KPIs.
 - Filters and Slicers: Implement interactive filters and slicers to allow users to explore data dynamically.
 - Drill-down functionality (optional but recommended): Enable users to drill down into more detailed data.
 - Insights: The dashboard should clearly present insights and answer at least 3-5 pre-defined business questions related to their theme.
- Deliverable: PowerBI Dashboard (.pbix file).

Phase 6: Final Project Report & Presentation (Day 3)

- Compile all deliverables.
- Prepare a comprehensive project presentation covering all aspects mentioned in the "Project Deliverables" section.
- Present their project findings, including a live demonstration of their database and PowerBI dashboard.

IV. Tools and Technologies

- Database Management System: MySQL
- Database Modeling & Administration: MySQL Workbench
- Business Intelligence Tool: Microsoft PowerBI Desktop
- ERD Modeling (Optional, but recommended): draw.io, Lucidchart, dbdiagram.io

- Text Editor for SQL: VS Code, Notepad++, Sublime Text

V. Assessment Criteria

Students will be assessed based on:

- Database Design: Correctness, efficiency, and normalization of the ERD and database schema.
- MySQL Implementation: Proper use of SQL for DDL and DML, data integrity, and completeness of the sample data.
- SQL Query Proficiency: Complexity, accuracy, and variety of SQL queries.
- PowerBI Dashboard Quality: Visual appeal, interactivity, clarity of insights, and effectiveness in answering business questions.
- Documentation & Presentation: Clarity, organization, completeness, and professionalism of the project report/presentation.
- Problem-Solving: Demonstrated ability to overcome technical challenges.
- Creativity & Originality: Uniqueness and depth of the chosen theme and analysis.

VI. Example Scenario: E-commerce Store Database

To illustrate, let's consider the "E-commerce Store Management" application.

Entities:

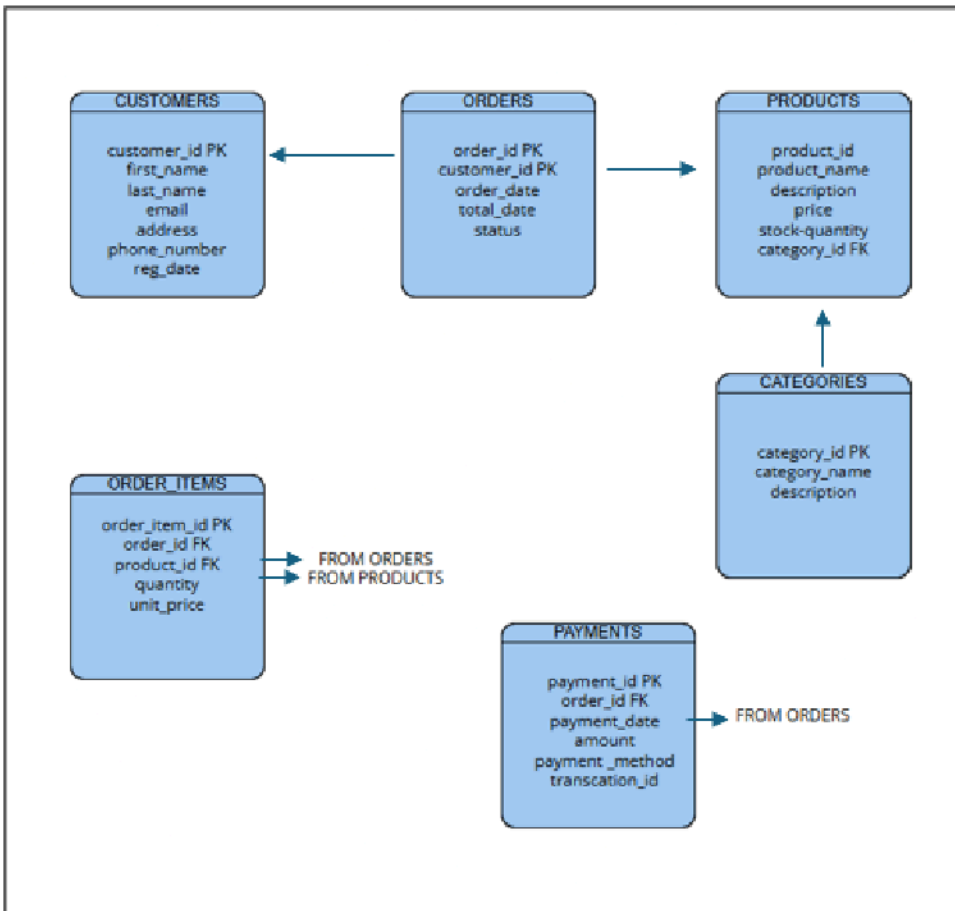
- **Customers:** customer_id (PK), first_name, last_name, email, address, phone_number, registration_date
- **Products:** product_id (PK), product_name, description, price, stock_quantity, category_id (FK)
- **Categories:** category_id (PK), category_name, description
- **Orders:** order_id (PK), customer_id (FK), order_date, total_amount, status (e.g., Pending, Shipped, Delivered)
- **Order_Items:** order_item_id (PK), order_id (FK), product_id (FK), quantity, unit_price
- **Payments:** payment_id (PK), order_id (FK), payment_date, amount, payment_method, transaction_id

Relationships:

- Customers have many Orders (One-to-Many)
- Orders contain many Products through Order_Items (Many-to-Many, resolved by Order_Items junction table)
- Products belong to one Category (Many-to-One)
- Orders have one Payment (One-to-One or One-to-Many depending on payment methods)

ERD Diagram: Conceptual Sketch

Students would create a detailed ERD diagram



MySQL Workbench:

Students would visually build this schema in MySQL Workbench, setting data types, primary keys, foreign keys, and relationships, which then allows for easy generation of the SQL script.

Queries (Examples):

1. INNER JOIN: Get customer names and their order details:

SQL

```
SELECT c.first_name, c.last_name, o.order_id, o.order_date, o.total_amount
FROM Customers c
INNER JOIN Orders o ON c.customer_id = o.customer_id;
```

2. LEFT JOIN & Aggregation: Total sales per product, including products with no sales:

SQL

```

SELECT p.product_name, c.category_name, SUM(oi.quantity * oi.unit_price) AS total_revenue
FROM Products p
LEFT JOIN Order_Items oi ON p.product_id = oi.product_id
LEFT JOIN Categories c ON p.category_id = c.category_id
GROUP BY p.product_name, c.category_name
ORDER BY total_revenue DESC;

```

3. Subquery: Find customers who have placed more than 5 orders:

```

SQL
SELECT c.first_name, c.last_name, COUNT(o.order_id) AS num_orders
FROM Customers c
JOIN Orders o ON c.customer_id = o.customer_id
GROUP BY c.customer_id
HAVING COUNT(o.order_id) > 5;

```

PowerBI Dashboard (Possible Visualizations):

- Total Sales by Month/Year (Line Chart): Track sales trends.
- Top 10 Selling Products (Bar Chart): Identify best-sellers.
- Sales by Category (Pie Chart): Understand revenue distribution across product categories.
- Customer Demographics (Cards/Tables): Number of active customers, average order value.
- Order Status Distribution (Donut Chart): Monitor order fulfillment.
- Geographical Sales (Map - if addresses are detailed enough): Visualize sales by region.
- Slicers: Date range, Product Category, Customer Name, Order Status.